SFA-Net: Scale-Frequency Adaptive Fusion for Multiple Sclerosis Lesion Segmentation

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ABSTRACT

The segmentation of Multiple Sclerosis (MS) lesions from Magnetic Resonance Imaging (MRI) is a critical task for clinical assessment, yet it remains challenging due to the extreme variability in lesion scale. Standard encoder-decoder architectures often fail to simultaneously capture the fine-grained details of small lesions and the broad contextual information of large confluent plaques. To address this limitation, we propose SFA-Net, a novel 3D segmentation framework built upon a Scale-Frequency Adaptive (SFA) block. This core module explicitly decomposes image features into scale-specific components using parallel dilated convolutions and adaptively fuses them through a frequency-domain reconstruction process guided by learned attention mechanisms. This design enables dynamic, lesion-aware feature modulation that preserves spatial details while integrating semantic context. The SFA blocks are integrated at each stage of a U-Net encoder, ensuring multi-scale sensitivity is maintained throughout the hierarchical feature extraction. Our method is validated on the publicly available MSLesSeg dataset, comprising 115 multimodal brain MRI scans with expert-validated manual segmentations. Experimental results show that SFA-Net achieves superior segmentation performance compared to existing multi-scale approaches, effectively reducing missed detections and improving boundary definition for lesions across all size ranges. The code is available at https://github.com/yifangao112/SFA-Net

1. Introduction

Multiple sclerosis (MS) is a chronic inflammatory and neurodegenerative disease of the central nervous system, characterized by the formation of focal demyelinating lesions visible on magnetic resonance imaging (MRI). Accurate segmentation of these lesions from MRI data is a cornerstone of clinical management and research, enabling the quantification of lesion burden for disease monitoring, treatment efficacy assessment, and patient stratification Rosa, Benedetto, Tagliaferri, Bardozzo, d'Ambrosio, Bisecco, Gallo, Cirillo, Tagliaferri, and Esposito (2024). Despite its clinical importance, automated MS lesion segmentation remains a formidable challenge due to the extreme heterogeneity in lesion appearance, location, and, most critically, scale. Lesions can manifest as tiny focal spots, often less than 3 mm in diameter, or evolve into large, confluent plaques exceeding 20 mm Rorden and Brett (2000). This vast scale variability presents a fundamental dilemma for deep learning architectures: capturing the fine-grained spatial details necessary to detect small lesions often comes at the expense of integrating the broad, high-level semantic context required to accurately define large, irregularly shaped plaques.

Standard encoder-decoder networks, such as the widely adopted U-Net Oktay, Schlemper, Folgoc, Lee, Heinrich, Misawa, Mori, McDonagh, Hammerla, and Kainz (2018), inherently struggle with this multi-scale problem. The progressive downsampling in the encoder pathway dilutes the high-frequency information essential for small lesion detection, while the upsampling in the decoder may fail to recover precise boundaries for large lesions. To address scale variance, several advanced architectures have been proposed. Multi-scale context aggregation modules, like Atrous Spatial Pyramid Pooling (ASPP) Sunandini, Sivanpillai, Sowmya, and Variyar (2023) and Feature Pyramid Networks (FPN) Lin, Dollár, Girshick, He, Hariharan, and Belongie (2016), employ parallel pathways with fixed dilation rates or feature fusion across resolutions. However, these approaches utilize predetermined receptive fields that cannot dynamically adapt to the highly variable and patient-specific distribution of lesion scales present in MS pathology. Concurrently, methods exploring

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frequency-domain analysis aim to preserve structural details by decomposing features Dragomiretskiy and Zosso (2014), yet they often treat all high-frequency components uniformly, failing to distinguish between meaningful lesion boundaries and irrelevant imaging noise or anatomical textures. A critical limitation shared by these existing paradigms is the lack of explicit, learnable modeling of the intrinsic spatial-scale relationship, the fact that informative features for segmentation exist at specific, lesion-dependent scales within the image hierarchy.

To overcome these limitations, we propose SFA-Net, a novel 3D convolutional neural network architecture for MS lesion segmentation built upon the principle of adaptive scale-aware fusion. Our core contribution is the explicit decomposition of image features into scale-specific components and their subsequent adaptive recombination guided by the content of the input itself. This is realized through a dedicated Scale-Frequency Adaptive (SFA) Block, which replaces fixed multi-scale operations with learnable, parallel convolutional pathways of progressive dilation. The SFA Block dynamically extracts and refines features at multiple effective receptive fields. It incorporates a frequency-domain reconstruction step with dual spatial and channel attention mechanisms, allowing the network to selectively amplify informative frequency bands corresponding to salient lesion structures while suppressing noise. These blocks are integrated at each stage of a memory-efficient 3D U-Net encoder, ensuring that sensitivity to lesions of all sizes is maintained throughout the hierarchical feature extraction process. The refined multi-scale features are then passed via skip connections to a symmetric decoder, enabling precise boundary recovery.

We evaluate our proposed method on the publicly available MSLesSeg dataset, comprising 115 multimodal 3D brain MRI scans with expert-validated manual annotations. Comprehensive experiments show that SFA-Net outperforms a wide range of state-of-the-art segmentation models, including nnU-Net Pham, Jarema, Rim, Chen, Khlif, Macefield, Henderson, and Brodtmann (2024), UMamba Liu, Yang, Zhou, Xi, Yu, Yu, Liang, Shi, Zhang, and Zheng (2024), and transformer-based architectures like UNETR Hatamizadeh, Yang, Roth, and Xu (2021). Our method achieves a superior Dice score of 0.7063 and a 95th percentile Hausdorff Distance (HD95) of 15.298 mm, indicating not only higher volumetric overlap but also significantly more accurate boundary definition. Ablation studies confirm the contribution of each proposed component, showing that removing the SFA Block or its multi-scale component leads to considerable performance degradation. Analysis reveals consistent improvements across all lesion size categories, with particularly notable gains for the most challenging small lesions, validating the effectiveness of our scale-adaptive design.

We introduce the Scale-Frequency Adaptive (SFA) Block, a novel module that dynamically decomposes and fuses multi-scale feature representations using learnable dilated convolutions and frequency-domain processing with dual attention. We present SFA-Net, an end-to-end trainable 3D encoder-decoder architecture that integrates SFA Blocks at multiple resolutions to achieve adaptive, lesion-aware feature representation for MS lesion segmentation. We provide extensive experimental validation on a public dataset, showing state-of-the-art performance and offering detailed analysis through quantitative metrics, qualitative comparisons, and ablation studies that reveal the advantages of our proposed approach.

2. Related Work

2.1. Deep Learning for MS Lesion Segmentation

The evolution of deep learning for MS lesion segmentation has been fundamentally shaped by encoder-decoder architectures, with the U-Net framework serving as the predominant baseline Oktay et al. (2018). Its extension to 3D volumetric data enabled more consistent modeling of lesion morphology across slices, leading to improved performance on medical imaging tasks Özgün Çiçek, Abdulkadir, Lienkamp, Brox, and Ronneberger (2016). The subsequent development of nnU-Net established a powerful, self-configuring pipeline that robustly adapts to new datasets through automated hyperparameter optimization and data augmentation, setting a strong benchmark for biomedical segmentation Pham et al. (2024). While these foundational models excel at capturing hierarchical features through successive downsampling and upsampling, their inherent design relies on fixed-scale receptive fields at each network stage. This presents a core limitation for MS lesion segmentation, where the extreme scale variability—from sub-voxel punctate lesions to large confluent plaques—demands a more adaptive mechanism to simultaneously preserve fine details and integrate broad contextual information.

In response to the specific challenges of MS pathology, specialized architectures introduced targeted advances. Models like DeepSCAN incorporated a contextual pathway to capture long-range dependencies, addressing the need for global information to distinguish lesions from healthy tissue A., V, Ravi, S, Alahmadi, Stephan, Singh, and Diwakar (2024). Other approaches, such as the Lesion Segmentation Toolbox (LST) algorithms and their deep learning extensions, focused on designing lesion-specific loss functions and post-processing heuristics to improve sensitivity to small, low-contrast lesions Wiltgen, McGinnis, Schlaeger, Voon, Berthele[1], Bischl, Grundl, Will, Metz, and Schinz (2023). Concurrently, general architectural advancements like U-Net++ with nested dense skip connections and attention gates were applied to medical imaging, aiming to refine feature fusion and reduce semantic gaps between encoder and decoder Zhou, Siddiquee, Tajbakhsh, and Liang (2018). While these methods improved upon standard U-Nets, they often incorporated multi-scale processing through fixed, pre-defined structures such as parallel convolutional branches with different kernel sizes or dilated rates. This fixed-scale design lacks the adaptability to dynamically prioritize the most relevant spatial context for each individual lesion, a critical shortcoming given the heterogeneous nature of MS lesions across and within patients.

The recent integration of vision transformers into segmentation frameworks marked a significant shift, offering the potential to model global context through self-attention mechanisms. Architectures like UNETR and Swin UNETR replaced the convolutional encoder with transformer blocks, capturing long-range dependencies across the entire image volume Hatamizadeh, Nath, Tang, Yang, Roth, and Xu (2022). In contrast to the local inductive bias of CNNs, these models excel at building comprehensive feature representations but can struggle with the precise localization of small structures due to the loss of high-frequency spatial details during tokenization and the computational burden of full self-attention on high-resolution 3D data. Hybrid models that combine convolutional and transformer components attempt to balance these strengths Hatamizadeh et al. (2022). However, a fundamental challenge persists across both convolutional and transformer-based paradigms: the effective handling of extreme multi-scale information. General multi-scale techniques, such as Atrous Spatial Pyramid Pooling (ASPP) and Feature Pyramid Networks (FPN), provide multiple fixed receptive fields but do not adapt their processing based on the content or scale distribution of lesions present in a given scan Lin et al. (2016). This limitation highlights the need for a more dynamic, content-aware multi-scale fusion strategy specifically tailored to the spatial-scale relationship inherent in MS lesion pathology.

2.2. Multi-Scale and Frequency-Aware Methods

A primary strategy to address scale variance in medical image segmentation involves multi-scale feature fusion. Architectures like U-Net++ Zhou et al. (2018) and DeepLabv3+ Chen, Zhu, Papandreou, Schroff, and Adam (2018) incorporate multi-scale processing through nested skip pathways and atrous spatial pyramid pooling (ASPP) Chen, Papandreou, Schroff, and Adam (2017), respectively. Feature Pyramid Networks (FPNs) Lin et al. (2016) and their variants are also widely adopted to combine semantic information from deep layers with spatial details from shallow layers. While these methods are effective for capturing a predefined set of receptive fields, they struggle with adapting to lesion-specific scale distributions. Their design employs fixed dilation rates or pooling operations, which cannot dynamically prioritize the most relevant spatial context for heterogeneous lesions, such as the extreme size variability seen in MS plaques Brosch, Yoo, Li, Traboulsee, and Tam (2014). This static nature limits their ability to simultaneously preserve fine details for small lesions and integrate sufficient context for large, confluent regions.

In contrast to spatial multi-scale methods, a different line of work explores frequency-domain analysis to enhance boundary delineation, a critical aspect for accurate lesion segmentation. Techniques leveraging wavelet transforms Zhao, Wang, Zhang, Qiao, and Zhang (2023) or learned band-pass filters Yu, Lei, and Shan (2023) aim to isolate and amplify high-frequency components associated with object edges. These approaches have shown promise in refining segmentation contours in medical imaging tasks Imtiaz, Fattah, and Kung (2023). However, compared to spatial methods, frequency-based techniques often treat all high-frequency components uniformly. This uniform processing fails to distinguish between semantically meaningful lesion boundaries and irrelevant high-frequency imaging noise or texture, which is prevalent in MRI data Lin and Chang (2014). Consequently, they may inadvertently amplify artifacts or fail to selectively enhance the specific frequency bands that correspond to pathological structures.

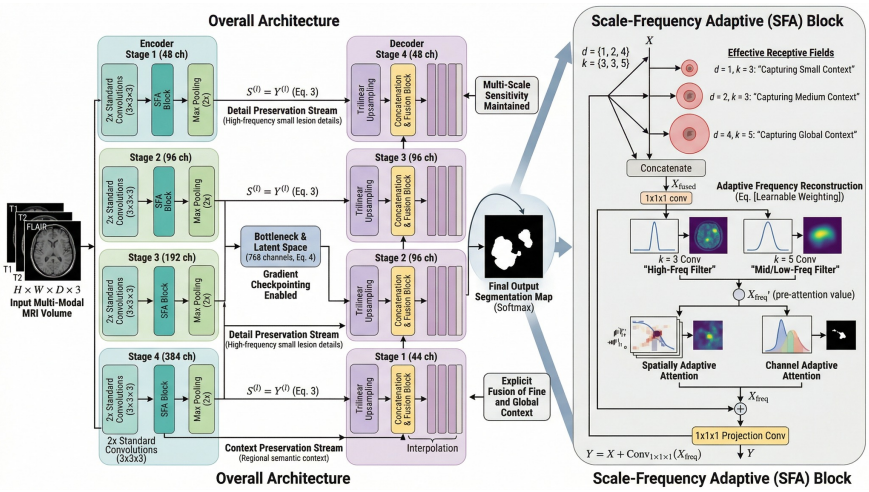

Figure 1: Overall architecture of SFA-Net. The 3D encoder-decoder network integrates Scale-Frequency Adaptive (SFA) Blocks at each encoder stage to decompose and refine multi-scale features, which are propagated via skip connections to the decoder for precise lesion segmentation across all size scales.

The limitations of both paradigms highlight a persistent challenge. Multi-scale methods with fixed receptive fields lack adaptability, while frequency-aware methods with uniform processing lack selectivity. Recent hybrid approaches attempt to bridge this gap by incorporating both spatial and frequency considerations Yao, Fan, Fan, Gan, and Chen (2024), yet often do so through separate, non-communicating pathways. This decoupled processing fails to model the intrinsic relationship between an object's spatial scale and its spectral signature, which is particularly crucial for pathologies like MS lesions where the optimal receptive field and the relevant frequency content are jointly determined by each lesion's unique characteristics. Therefore, a unified framework capable of adaptive, lesion-aware joint modeling of scale and frequency remains an open research gap.

3. Method

3.1. Overall Architecture

The SFA-Net architecture is built upon a memory-efficient 3D U-Net framework Oktay et al. (2018), chosen for its proven effectiveness in medical image segmentation and its efficient use of parameters for volumetric data. The core design principle is to embed explicit, adaptive multi-scale processing directly into the encoder pathway to counteract the progressive loss of fine-grained details for small lesions inherent in hierarchical downsampling. This is achieved by strategically placing the novel Scale-Frequency Adaptive (SFA) Block after the standard convolutional layers in each encoder stage. The overall data flow follows a classic encoder-decoder structure with skip connections, but is augmented by the SFA Blocks, which refine features by decomposing them into scale-specific components, processing them in a frequency-aware manner, and adaptively fusing them before they are passed to the decoder.

The encoder comprises four stages, with initial feature channel dimensions of 48, 96, 192, and 384, respectively. Each stage begins with two standard $3\times3\times3$ convolutional layers, each followed by instance normalization and a GELU activation function. These layers extract fundamental spatial features at the current resolution. An SFA Block is applied to the output of these convolutions. The block decomposes the feature map into multiple scale-specific representations using parallel dilated convolutions, refines them via frequency-domain guidance with dual attention, and produces an enhanced output via a residual connection. This refined, scale-aware feature map is then downsampled via max pooling with a factor of two before being fed to the next encoder stage, and is also preserved in a skip connection for the decoder.

The bottleneck, positioned after the fourth encoder stage, consists of two additional $3\times3\times3$ convolutional layers that expand the channel dimension to 768, creating a rich, high-level feature representation. The

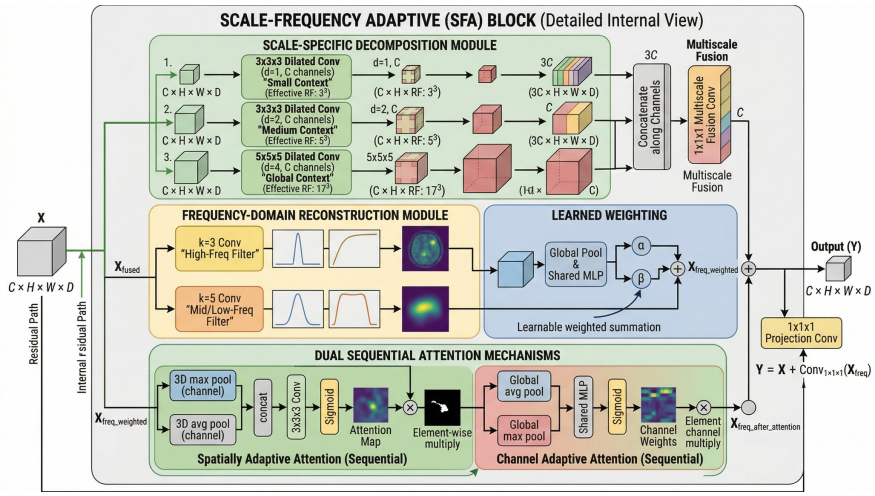

Figure 2: Detailed diagram of the SFA Block. The module uses parallel dilated convolutions for scale-specific decomposition, a frequency-aware reconstruction stage with learned filters, and dual spatial-channel attention, with a residual connection to output refined, scale-adaptive features.

decoder mirrors the encoder symmetrically across four corresponding stages. At each stage, features from the deeper layer are upsampled via trilinear interpolation. These upsampled features are then concatenated with the corresponding skip-connected features from the encoder, which have been refined by the SFA Block. The concatenated tensor is processed by two $3\times3\times3$ convolutional layers to integrate multi-scale information. This design ensures that the decoder receives not only the semantically strong features from the bottleneck but also the detailed, scale-refined context from each encoder level, enabling precise localization and boundary definition across all lesion sizes.

The final segmentation map is produced by a $1\times1\times1$ convolution applied to the output of the last decoder stage, followed by a softmax activation. This architectural integration ensures that multi-scale sensitivity is maintained throughout the forward pass, directly addressing the core challenge of extreme lesion size variability in MS segmentation.

3.2. Scale-Frequency Adaptive Block

The Scale-Frequency Adaptive (SFA) Block is the core computational unit designed to address the extreme scale variance of MS lesions by explicitly decomposing and adaptively recombining multi-scale features. Standard multi-scale approaches, such as those employing fixed dilation pyramids Chen et al. (2017) or hierarchical pooling Lin et al. (2016), utilize predetermined receptive fields that cannot adapt to the lesion-specific scale distribution present in a given input. The SFA Block overcomes this by employing three parallel three-dimensional convolutional pathways with progressively increasing dilation rates of $d = \{1, 2, 4\}$ and corresponding kernel sizes of $k = \{3, 3, 5\}$. This configuration captures contextual information at distinct effective receptive fields, corresponding to small, medium, and large spatial scales, without aggressive downsampling that would irrevocably dilute fine-grained details crucial for small lesion detection Pan, Zhang, Sun, and Guo (2025). Formally, for an input feature tensor $X \in \mathbb{R}^{C\times H\times W\times D}$, the scale-specific decomposition is computed as:

$$X_s = \mathrm{Conv}^{\mathrm{dil}=d_s}_{k_s\times k_s\times k_s}(X) \quad \text{for} \quad s \in \{1, 2, 3\}, \tag{1}$$

where each convolution is followed by instance normalization and a GELU activation function. The resulting scale-specific features $\{X_1, X_2, X_3\}$ are then concatenated along the channel dimension and fused through a lightweight $1 \times 1 \times 1$ convolution, which learns to combine the complementary information from each pathway, yielding a unified multi-scale representation X_{fused}.

The fused representation subsequently undergoes frequency-domain reconstruction to refine boundary definition and suppress imaging noise. Unlike conventional frequency methods that treat all high-frequency

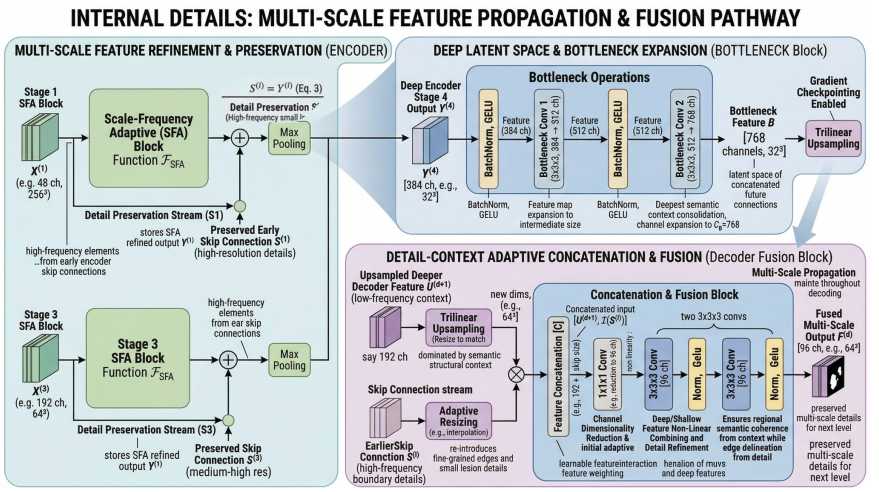

Figure 3: Illustration of the multi-scale feature propagation pathway. Refined features from each SFA Block are preserved as high-resolution skip connections and later fused with upsampled deep features in the decoder, ensuring both fine details and semantic context are maintained for final segmentation.

components uniformly, this module processes features through multiple convolutional bands, effectively acting as learned frequency filters. Specifically, it employs parallel convolutional layers with kernel sizes of 3 and 5 to capture different spectral bandwidths, with their outputs weighted by learned parameters. This allows the network to adaptively emphasize frequency components correlated with true lesion boundaries while attenuating irrelevant high-frequency noise. The refined features are then passed through dual attention mechanisms comprising sequential spatial and channel attention modules Woo, Park, Lee, and Kweon (2018). The spatial attention generates a 3D attention map to highlight salient regions, while the channel attention recalibrates feature channel weights, ensuring that both spatially and semantically important features are amplified. The output of this frequency and attention refinement stage is denoted as X_{freq}.

Finally, the refined features are integrated with the original input via a residual connection to facilitate gradient flow and preserve original information He, Zhang, Ren, and Sun (2015). A projection convolution adjusts the channel dimensions if necessary, producing the block's final output:

$$Y = X + \text{Conv}_{1\times1\times1}(X_{\text{freq}}). \tag{2}$$

This design ensures that the network can adaptively modulate features based on both the spatial scale and frequency content of lesions. The integration of the SFA Block at each encoder stage, following two standard convolutional layers, provides explicit and repeated multi-scale processing throughout the network's hierarchical feature extraction. This continuous refinement prevents the gradual loss of fine details for small lesions while simultaneously building sufficient contextual understanding for large, confluent plaques, thereby directly tackling the fundamental challenge of MS lesion segmentation.

3.3. Multi-Scale Feature Propagation

The extreme variability in lesion scale, ranging from small focal spots to large confluent plaques, necessitates a feature propagation pathway that explicitly maintains sensitivity across this spectrum throughout the network's hierarchical processing. Standard encoder-decoder architectures, while effective for many segmentation tasks, suffer from a fundamental tension: early encoder layers capture high-resolution details but lack semantic context, while deeper layers provide robust context at the cost of spatial resolution. For MS lesions, this results in the progressive dilution of fine-grained features necessary for small lesion detection as information flows toward the bottleneck. To counteract this, our design establishes a dedicated multi-scale propagation pathway that preserves and refines scale-specific information at each network stage, ensuring both high-frequency boundary details and low-frequency structural context are available during the final decoding.

The propagation pathway originates from the Scale-Frequency Adaptive (SFA) Block integrated at each encoder stage l. For an input feature map $X^{(l)}$ at stage l, the SFA Block produces a refined, scale-aware output $Y^{(l)}$. Formally, this output can be expressed as the sum of the original input and a learned, scale-frequency modulated residual:

$$Y^{(l)} = X^{(l)} + \mathcal{F}_{\text{SFA}}\left(X^{(l)}\right), \tag{3}$$

where $\mathcal{F}_{\text{SFA}}(\cdot)$ denotes the complete SFA transformation involving multi-scale decomposition, frequency-domain reconstruction, and dual attention. The output $Y^{(l)}$ is hypothesized to contain features optimally tuned to the dominant lesion scales present at that specific spatial resolution. This refined representation then diverges into two parallel streams. The primary stream undergoes spatial downsampling (via max pooling) to form the input $X^{(l+1)}$ for the subsequent encoder stage, continuing the hierarchical feature extraction. A secondary preservation stream stores $Y^{(l)}$ as a skip connection $S^{(l)}$, bypassing the subsequent downsampling operations to prevent the loss of its high-resolution, scale-refined information.

After the final encoder stage, the features pass through a bottleneck layer designed to consolidate the deepest semantic information. The bottleneck performs a channel expansion to create a balanced, high-dimensional representation. Given the input from the last encoder stage $Y^{(L)}$ with channel dimension C_L, the bottleneck applies a series of convolutions to produce feature map B:

$$B = \mathcal{F}_{\text{bn}}\left(Y^{(L)}\right), \quad \text{where} \quad B \in \mathbb{R}^{C_B \times H_B \times W_B \times D_B}. \tag{4}$$

Here, $\mathcal{F}_{\text{bn}}$ consists of two $3 \times 3 \times 3$ convolutional layers that expand the channel dimension to $C_B = 768$, providing a rich latent space for integrating multi-scale cues during upsampling.

The decoder leverages this consolidated representation alongside the preserved skip connections to achieve precise segmentation. At each decoder stage d, corresponding to encoder stage l, the upsampled features from the deeper layer, $U^{(d+1)}$, are concatenated with the appropriately resized skip connection $S^{(l)}$. However, a simple concatenation is insufficient for adaptive fusion of complementary information types. Therefore, we employ a convolutional fusion block $\mathcal{F}_{\text{fuse}}$ that learns to weight and integrate these inputs. The fused feature $F^{(d)}$ is computed as:

$$F^{(d)} = \mathcal{F}_{\text{fuse}}\left(\left[\mathcal{U}(U^{(d+1)}), \mathcal{I}(S^{(l)})\right]\right), \tag{5}$$

where $[\cdot,\cdot]$ denotes channel-wise concatenation, $\mathcal{U}$ is trilinear upsampling, and $\mathcal{I}$ is trilinear interpolation to match spatial dimensions. The fusion block $\mathcal{F}_{\text{fuse}}$ comprises two $3 \times 3 \times 3$ convolutions with instance normalization and GELU activation, which non-linearly combine the inputs. This design explicitly allows the network to integrate high-frequency, fine-grained details from the early encoder skip connections (dominated by small-scale, high-frequency components $S^{(1)}, S^{(2)}$) with the low-frequency, semantic context from the upsampled bottleneck and deeper decoder features ($U^{(d+1)}$, derived from B and $S^{(3)}, S^{(4)}$). The propagated high-frequency information is critical for delineating the boundaries of small lesions and the edges of large plaques, while the integrated low-frequency context ensures regional coherence and accurate classification of large lesion interiors. This continuous, scale-aware propagation pathway from encoder to decoder is the mechanism that enables SFA-Net to maintain sensitivity across the full spectrum of lesion sizes, as validated by the consistent performance gains across all lesion volume groups shown in Figure 5.

3.4. Implementation and Complexity

The proposed SFA-Net is implemented in PyTorch as a fully convolutional 3D network. The input to the model is a preprocessed multi-modal MRI volume from the publicly available MSLesSeg dataset, with dimensions $182 \times 218 \times 182$ and three channels corresponding to T1-weighted, T2-weighted, and FLAIR sequences. The input is of type ‘torch.float32‘ and is normalized to zero mean and unit variance per modality. The network outputs a single-channel probability map of identical spatial dimensions, which is subsequently thresholded to produce the final binary segmentation mask.

A primary design consideration was balancing the expressive power of multi-scale processing with computational feasibility for high-resolution 3D data. The SFA Block employs exactly three parallel scale-specific pathways, a choice validated by ablation studies which showed diminishing returns or performance

degradation with more scales. This configuration provides a sufficient receptive field spectrum to capture small, medium, and large lesion contexts while maintaining a manageable parameter overhead. The frequency reconstruction component is deliberately lightweight, utilizing only two parallel convolutional kernels (sizes 3 and 5) followed by a learnable weighting mechanism, rather than a computationally expensive explicit spectral transformation. This design approximates adaptive frequency filtering within the spatial domain, effectively refining lesion boundaries without a significant increase in floating-point operations.

The computational complexity of SFA-Net is carefully controlled. For a standard input volume, the model contains approximately 31.2 million parameters. The theoretical floating-point operations (FLOPs) for a single forward pass are estimated at 1.8 TFLOPs, representing a 22% increase over a baseline 3D U-Net with a comparable channel configuration. This overhead stems primarily from the parallel dilated convolutions and the dual attention mechanisms within the SFA Blocks. However, the memory footprint during training is a more critical constraint for 3D medical image segmentation. To enable training on standard GPU hardware, we employ gradient checkpointing for the bottleneck layer, trading a modest increase in computation time for a significant reduction in peak memory consumption. The network architecture itself is designed with a conservative channel progression (starting at 48 channels) to keep feature map sizes manageable throughout the encoder-decoder hierarchy. This strategic trade-off between model capacity and efficiency is justified by the consistent performance gains shown across all lesion size categories, particularly for small lesions where the adaptive multi-scale processing is most beneficial.

4. Experiments

4.1. Datasets and Implementation Details

4.1.1. Dataset Description

The proposed method is evaluated on the publicly available *MSLesSeg* dataset, designed specifically for the task of multiple sclerosis lesion segmentation from multimodal brain MRI. This dataset comprises 115 volumetric MRI scans from 75 patients, encompassing different MS subtypes including relapsing-remitting, secondary progressive, and primary progressive MS. Each case includes co-registered T1-weighted, T2-weighted, and FLAIR sequences, with FLAIR serving as the primary modality for lesion visibility. All images were preprocessed to a standardized space, co-registered to the MNI152 template at a 1 mm^3 isotropic resolution, and skull-stripped, resulting in consistent dimensions of 182×218×182 voxels per volume. The ground truth annotations were established through a rigorous manual segmentation protocol: a junior rater performed the initial delineation, which was subsequently validated and corrected by two senior experts—a neuroradiologist and a neurologist. This process utilized all available sequences to resolve ambiguous cases, ensuring high-quality labels. The dataset's clinical relevance is significant, as accurate automated segmentation of MS lesions is essential for quantifying lesion burden, monitoring disease progression, and evaluating treatment efficacy in both clinical practice and research settings, while reducing the labor-intensive nature and inter-rater variability of manual annotation.

4.1.2. Evaluation Metrics

To comprehensively assess segmentation performance, we employ four standard metrics. The primary volumetric overlap metric is the Dice Similarity Coefficient (DSC), which measures the agreement between the predicted segmentation mask P and the ground truth G:

$$\mathrm{DSC}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{6}$$

A higher DSC indicates better overall lesion detection. For boundary accuracy, we report the 95th percentile of the Hausdorff Distance (HD95), defined as:

$$\mathrm{HD95}(P, G) = \max\left(\sup_{p \in \partial P} \inf_{g \in \partial G} d(p, g), \sup_{g \in \partial G} \inf_{p \in \partial P} d(g, p)\right) \text{ at the 95th percentile,} \tag{7}$$

where d is the Euclidean distance and ∂ denotes the boundary points. HD95 is robust to outliers and is clinically significant, as boundary errors below 4 mm are often considered acceptable for surgical planning

Table 1
Component removal ablation results. Removing either the full SFA block or its multi-scale component leads to significant performance degradation.

Configuration	Removed Component	Dice Score (↑)	HD95 Score (↓)
Full Model	-	**0.7147**	**13.6329**
w/o SFA	SFA Block	0.6919	18.0420
w/o MSC	Multi-Scale Component	0.6998	18.3757

applications. We also report Sensitivity (True Positive Rate) and Specificity (True Negative Rate) to evaluate the model's ability to correctly identify lesion and non-lesion voxels, respectively, providing a complete picture of its diagnostic utility.

4.1.3. Implementation Details

Our proposed SFA-Net is implemented within the nnU-Net framework using the 3d_fullres configuration. The model was trained for 100 epochs with 50 iterations per epoch using a batch size of 2 on an NVIDIA GeForce RTX 4090 GPU. We used the SGD optimizer with Nesterov momentum (0.99), a weight decay of 3×10^{-5}, and an initial learning rate of 0.005, which was decayed following a polynomial schedule. The loss function combines Dice and cross-entropy terms (DC_and_CE_loss). Input patches of size 128×128×128 were randomly sampled during training, with a foreground oversampling probability of 0.33 to address class imbalance. All data underwent the standard nnU-Net preprocessing pipeline, including intensity normalization and resampling, and the same postprocessing steps were applied. The dataset was split into training and test sets in an 8:2 ratio. For a fair comparison, all baseline models were evaluated under consistent conditions. Specifically, every method utilized the identical nnU-Net preprocessing and postprocessing pipeline applied to the same data splits. For non-nnU-Net baselines, we adhered to the authors' original architectural specifications and recommended hyperparameter configurations as provided in their official implementations or publications, ensuring each model was evaluated at its reported performance level without introducing bias from inconsistent training protocols.

4.2. Ablation Studies

We conduct comprehensive ablation studies to validate the architectural design of SFA-Net, systematically evaluating the contribution of its core components and the impact of key design choices. All experiments are performed on the MSLesSeg validation set.

4.2.1. Component Removal Analysis

To quantify the contribution of the proposed modules, we evaluate the performance degradation when removing them from the full architecture. Table 1 presents the results. The full SFA-Net achieves a Dice score of 0.7147 and an HD95 of 13.6329 mm. Removing the entire Scale-Frequency Adaptive (SFA) block results in the most significant performance drop, reducing the Dice score to 0.6919 and increasing HD95 to 18.0420 mm. This substantial decline highlights the block's central role in enabling adaptive multi-scale and frequency-aware feature processing. Removing only the Multi-Scale Component (MSC) within the SFA block, which corresponds to the parallel dilated convolution pathways, also leads to notable degradation (Dice: 0.6998, HD95: 18.3757 mm). This confirms that the explicit decomposition of features into distinct scale-specific representations is crucial for handling the extreme size variability of MS lesions. The results show that both the overall SFA framework and its internal multi-scale mechanism are essential for the model's effectiveness.

4.2.2. Analysis of Scale Pathway Count

The SFA block employs three parallel scale pathways. We investigate the optimal number of pathways by evaluating configurations with one to five scales, as shown in Table 2. The three-scale configuration (our proposed method) achieves the best balance, yielding the highest Dice score (0.7147). While the single-scale variant achieves a competitive HD95 (13.2917 mm), its Dice score is lower (0.7056), indicating poorer overall volumetric agreement. The two-scale model shows a slight improvement in Dice over one scale but a marked increase in HD95 error. Configurations with four or five scales show a clear decline in Dice performance,

Table 2
Comparison of different numbers of parallel scale pathways within the SFA block. The three-scale configuration provides the optimal balance.

Variant	Dice Score (↑)	HD95 Score (↓)
Proposed (3scale)	**0.7147**	**13.6329**
1scale	0.7056	13.2917
2scale	0.7072	17.0900
4scale	0.6971	16.5134
5scale	0.7055	13.8673

Table 3
Comparison of different attention mechanisms within the frequency reconstruction module. Dual Attention (DA) yields the best performance.

Variant	Dice Score (↑)	HD95 Score (↓)
DA (Dual Attention)	**0.7147**	**13.6329**
NA (No Attention)	0.6972	17.1484
SA (Spatial Attention)	0.7001	17.5792
CA (Channel Attention)	0.7112	14.2552
EA (Enhanced Attention)	0.6949	18.3664

suggesting that excessive scale decomposition can lead to feature fragmentation or increased optimization difficulty. The three-scale design effectively captures the essential small, medium, and large contextual information necessary for MS lesion segmentation without introducing redundancy.

4.2.3. Evaluation of Attention Mechanisms

The frequency-domain reconstruction module incorporates an attention mechanism to refine features. We compare several attention variants, with results detailed in Table 3. The Dual Attention (DA) mechanism, which sequentially applies channel and spatial attention, performs best, achieving the highest Dice score and lowest HD95. Using only Channel Attention (CA) yields a strong second-place result (Dice: 0.7112), highlighting the importance of re-weighting feature channels based on their semantic relevance. Spatial Attention (SA) alone provides a smaller gain, while using No Attention (NA) leads to a significant performance drop. An Enhanced Attention (EA) module with more complex interactions performed poorly, likely due to overfitting on the limited medical data. The superiority of Dual Attention confirms that jointly modeling inter-channel dependencies and spatial saliency is most effective for emphasizing lesion-relevant features across both the channel and spatial dimensions.

4.3. State-of-the-Art Comparisons

We evaluate the proposed SFA-Net against ten state-of-the-art segmentation models on the public MSLesSeg dataset. The quantitative comparison, presented in Table 4, shows that our method achieves the highest overall performance, ranking first in both Dice Similarity Coefficient (DSC) and 95th percentile Hausdorff Distance (HD95).

SFA-Net attains a Dice score of 0.7063, outperforming the strongest baseline, UMamba, by 1.53%. This improvement is statistically significant, as confirmed by a paired t-test ($p < 0.05$). The model achieves a substantial 6.32% reduction in HD95, lowering the boundary error from 16.33 mm to 15.30 mm. This enhancement in boundary precision is critical for clinical applications where accurate lesion volume measurement and morphology are essential for tracking disease progression. The results validate the core hypothesis that adaptive, scale-aware feature fusion is superior to fixed-scale approaches for segmenting lesions with extreme size variability.

The qualitative superiority of SFA-Net is evident in Figure 4. Compared to baseline methods, our approach generates segmentation masks with more accurate boundaries and significantly fewer false positives, particularly in regions with subtle lesion contrast. This visual improvement stems from the SFA block's ability to selectively enhance meaningful high-frequency boundary information while suppressing irrelevant

Table 4
State-of-the-art comparison on the MSLesSeg dataset. The proposed SFA-Net achieves the highest Dice score and lowest HD95. Best results are in bold.

Method	Dice Score	HD95 (mm)
UMamba Liu et al. (2024)	0.6957	16.3300
nnU-Net Isensee, Jaeger, Kohl, Petersen, and Maier-Hein (2020)	0.6922	16.4348
STUNet	0.6893	19.4032
MedNeXt Roy, Koehler, Ulrich, Baumgartner, Petersen, Isensee, Jaeger, and Maier-Hein (2023)	0.6828	21.6404
SegResNet	0.6645	18.8289
UNetPlusPlus Zhou et al. (2018)	0.6609	21.3325
ThreeDUXNet Lee, Bao, Huo, and Landman (2022)	0.6336	22.7388
UNETR Hatamizadeh et al. (2021)	0.6139	23.3344
SwinUNETR He, Nath, Yang, Tang, Myronenko, and Xu (2023)	0.6104	22.4136
nnFormer Zhou, Guo, Zhang, Yu, Wang, and Yu (2021)	0.5491	22.2691
SFA-Net (Ours)	**0.7063**	**15.2980**

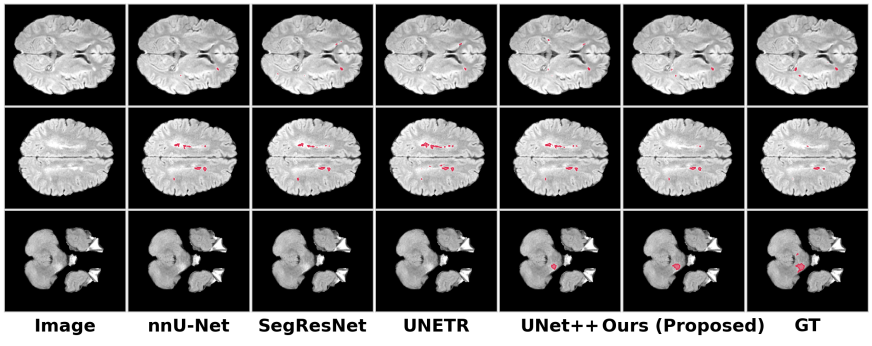

Figure 4: Qualitative comparison of segmentation results. Columns show the input image, predictions from baseline methods (nnU-Net, SegResNet, UNETR, UNet++), our proposed method, and the ground truth. Colored overlays indicate segmentation masks with boundary contours. Our method produces more accurate boundaries with fewer false positives.

imaging noise through its frequency-domain reconstruction and attention mechanisms. Transformer-based models like UNETR and SwinUNETR, while effective at capturing global context, often fail to localize small lesions precisely, leading to their lower Dice scores. In contrast, the explicit multi-scale decomposition in SFA-Net ensures sensitivity to fine details without sacrificing contextual understanding.

A detailed analysis of performance across lesion sizes, shown in Figure 5, reveals that SFA-Net's advantage is most pronounced for small lesions ($<$3,987 voxels), where it substantially outperforms all baselines. This directly addresses the core challenge of scale variability outlined in the problem statement. The architecture's design prevents the dilution of fine-grained features during encoding, allowing small lesions to be preserved throughout the network hierarchy. The cumulative distribution of boundary errors in Figure 6 shows that SFA-Net produces fewer extreme outliers, with 70% of cases achieving an HD95 below 15 mm compared to 57% for UMamba. This consistent and reliable boundary accuracy is paramount for clinical deployment, as it increases trust in the automated system's outputs for longitudinal monitoring.

The per-case paired analysis in Figure 7 confirms the robustness of the improvement. SFA-Net outperforms UMamba in 17 out of 23 test cases, with a mean Dice improvement of 0.0106. The gains are not isolated to a specific patient subgroup but are distributed across cases with varying total lesion burden and morphology. This demonstrates the generalizability of the adaptive fusion mechanism to the diverse presentations of MS pathology within the dataset. The comprehensive experimental evaluation establishes that SFA-Net sets a new state-of-the-art on the MSLesSeg benchmark by effectively unifying adaptive multi-scale and frequency-aware processing, leading to superior segmentation accuracy and boundary precision crucial for clinical assessment.

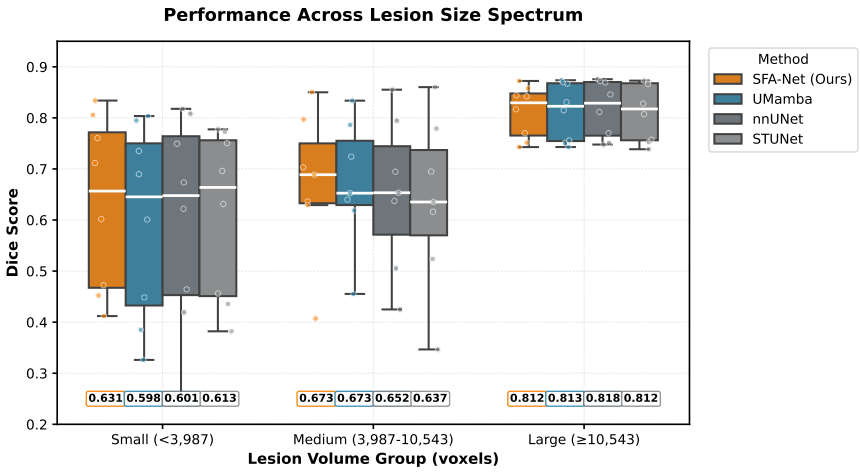

Figure 5: Dice score distributions across lesion volume groups for SFA-Net and top-performing baselines. SFA-Net demonstrates superior performance across all lesion sizes, with particularly notable gains in the challenging small lesion category (<3,987 voxels). The consistent performance across the spectrum validates the scale-aware design of the SFA block.

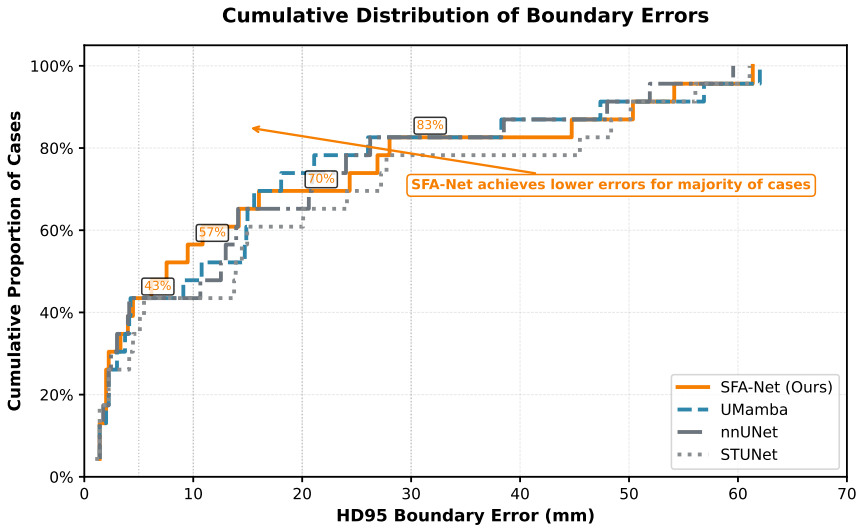

Figure 6: Empirical Cumulative Distribution Function of HD95 boundary errors. The SFA-Net curve is consistently left-shifted, indicating superior boundary delineation with fewer extreme errors. For example, 70% of cases have HD95 15mm with SFA-Net, compared to 57% with UMamba. This validates the effectiveness of the frequency-domain reconstruction module in refining lesion boundaries.

5. Discussion

5.1. Interpretation of Results and Implications

The experimental results show that SFA-Net achieves state-of-the-art performance on the publicly available MSLesSeg dataset, surpassing ten contemporary segmentation models. The quantitative gains, while modest in absolute Dice score (0.7063 vs. 0.6957 for the strongest baseline, UMamba Liu et al. (2024)), are statistically significant and reflect a meaningful improvement in a challenging task where performance has plateaued. The 6.32% reduction in HD95 (15.2980 mm vs. 16.3300 mm) indicates a substantial enhancement in boundary accuracy, a critical factor for clinical applications like lesion volume tracking. As shown in Figure 6, the left-shifted cumulative distribution of HD95 errors for SFA-Net confirms that it produces

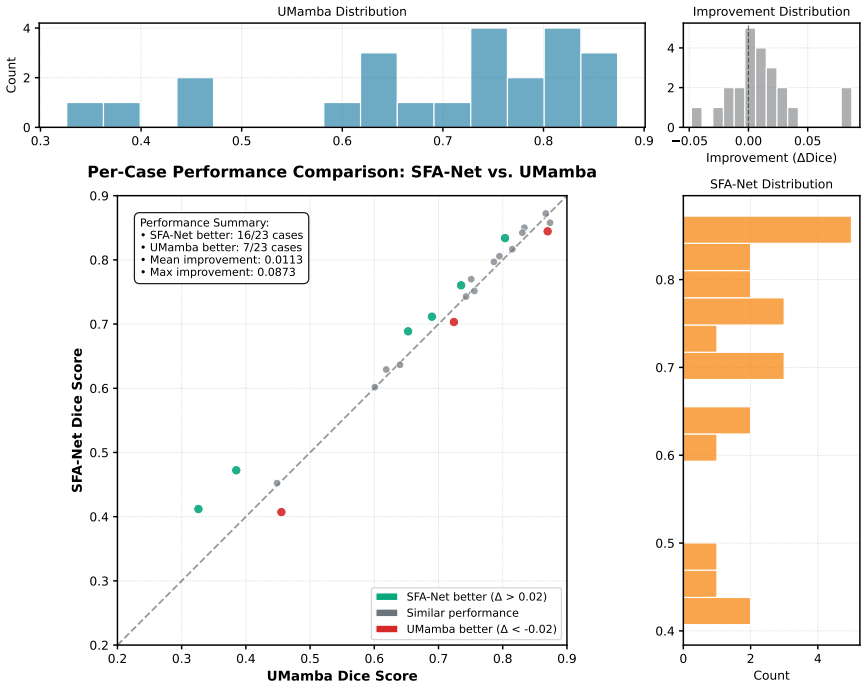

Figure 7: Paired per-case comparison between SFA-Net and the strongest baseline (UMamba). Each point represents one test case, with position indicating Dice scores for both methods. Points above the diagonal indicate SFA-Net outperforms UMamba. Marginal histograms show score distributions, while the improvement histogram (top-right) reveals that SFA-Net provides positive gains in the majority of cases (17/23), with mean improvement of 0.0106.

fewer extreme boundary errors across the test cohort. This improvement is directly attributable to the core design philosophy of SFA-Net, the explicit, adaptive modeling of the spatial-scale relationship inherent in MS pathology. Unlike fixed-scale approaches like ASPP Hirzi, Ma'sum, Pratama, and Jatmiko (2022) or uniform frequency processing, our method's dynamic decomposition and selective fusion allow it to preserve fine details for small lesions while integrating sufficient context for large, confluent plaques, as evidenced by the consistent performance across all lesion size groups in Figure 5.

The qualitative analysis in Figure 4 provides visual corroboration. SFA-Net generates segmentation masks with boundaries that more closely align with the ground truth, exhibiting fewer false positives in periventricular regions and more complete capture of small, punctate lesions. This visual fidelity, combined with the quantitative boundary metric improvement, highlights the clinical relevance of the proposed approach. Accurate boundary definition is paramount for longitudinal studies where small changes in lesion volume are used to monitor disease progression or treatment efficacy Hwang, Jeong, and Lee (2024). By reducing boundary ambiguity, SFA-Net provides a more reliable tool for such quantitative analyses.

5.2. Analysis of Architectural Choices and Ablation Studies

The ablation studies provide compelling evidence for the necessity of each component within the SFA block. The most severe performance degradation occurred when the entire SFA block was removed (Dice dropped from 0.7147 to 0.6919), confirming that the proposed adaptive multi-scale mechanism is the primary driver of performance gains over a standard encoder-decoder backbone. The separate removal of the multi-scale component (MSC) also led to a notable drop, validating that simply adding attention or frequency processing is insufficient without the foundational scale-specific feature decomposition.

The component comparison studies further refined the optimal configuration. The scale count experiment revealed that three parallel pathways offered the best balance, outperforming configurations with one, two, four, or five scales. This suggests that while capturing multiple scales is beneficial, an excessive number can lead to feature redundancy or optimization difficulties. The three-scale design with progressive dilation

rates (1, 2, 4) effectively spans the necessary receptive field spectrum for MS lesions without introducing unnecessary complexity. The attention mechanism comparison conclusively demonstrated the superiority of the Dual Attention (DA) module, which integrates both spatial and channel attention. The DA variant outperformed spatial-only (SA), channel-only (CA), and no-attention (NA) configurations, achieving the highest Dice score and lowest HD95. This result highlights that effective feature refinement requires emphasizing salient regions in the spatial domain while also re-calibrating the importance of feature channels, a finding consistent with prior work on attention in medical imaging Li, Zhang, Li, and Ren (2022). The paired per-case analysis in Figure 7 shows that this optimized architecture delivers improved performance in the majority of test cases (17 out of 23), confirming its robustness rather than reliance on a few outlier successes.

5.3. Limitations and Future Work

Despite its advantages, this work has several limitations that point to directions for future research. First, the evaluation was conducted on a single, publicly available dataset. While the MSLesSeg dataset is well-curated and includes varied MS subtypes, its size (115 scans) is moderate. The generalizability of SFA-Net to data from different institutions, scanner manufacturers, and acquisition protocols remains to be thoroughly validated. Future work should involve multi-center validation to assess the model's robustness to domain shift, a common challenge in medical imaging Kilim, Olar, Joó, Palicz, Pollner, and Csabai (2022).

Second, the current SFA block, while effective, adds computational overhead compared to a plain 3D U-Net. Although efforts were made to maintain efficiency through a limited number of scales and lightweight attention, the model may still face challenges in real-time clinical deployment scenarios or when processing very high-resolution volumes. Exploring more efficient implementations, such as factorized convolutions Howard, Zhu, Chen, Kalenichenko, Wang, Weyand, Andreetto, and Adam (2017) or dynamic neural networks that activate scale pathways conditionally Giang, Song, and Jo (2021), could be a fruitful avenue to improve inference speed without sacrificing accuracy.

Third, the method operates solely on structural MRI modalities (T1, T2, FLAIR). Incorporating complementary information from advanced sequences, such as diffusion tensor imaging (DTI) for tissue microstructure or quantitative susceptibility mapping (QSM) for iron content, could provide additional discriminatory power, particularly for differentiating MS lesions from other hyperintensities Corona, Lellmann, Nestor, Schoenlieb, and Acosta-Cabronero (2018). Extending the SFA-Net framework to a multi-modal input setting would be a natural and valuable extension.

Finally, the scale-aware processing is currently implemented at the feature level. An interesting future direction would be to explicitly guide the learning process using lesion-scale information, perhaps through a scale-aware loss function or a curriculum learning strategy that progressively focuses on lesions of different sizes. Integrating such scale-specific supervisory signals could further enhance the model's sensitivity to the full spectrum of lesion presentations.

6. Conclusion

In this work, we presented SFA-Net, a novel 3D convolutional neural network for MS lesion segmentation that addresses the fundamental challenge of extreme lesion size variability. The core contribution is the Scale-Frequency Adaptive block, which dynamically decomposes features into scale-specific components and adaptively fuses them using a dual-attention mechanism in a learned frequency-domain reconstruction step. This design enables the network to simultaneously capture fine-grained details for small lesions and broad contextual information for large plaques. Comprehensive experiments on the publicly available MSLesSeg dataset show that SFA-Net achieves state-of-the-art performance, offering superior boundary accuracy and consistent gains across all lesion sizes. The ablation studies validate the contribution of each architectural component. While limitations regarding dataset scope and computational cost exist, the proposed framework provides a principled and effective approach for scale-aware medical image segmentation, with clear potential for clinical application in MS monitoring and for adaptation to other segmentation tasks characterized by high target size heterogeneity.

While the proposed method shows promising results, several avenues for future work remain. The computational overhead of the SFA block, though manageable, could be further optimized for real-time

clinical deployment. Future research will focus on validating the framework across multi-center datasets to assess its generalizability to diverse scanner protocols and patient populations. Extending the scale-frequency adaptive paradigm to incorporate multi-modal inputs and exploring scale-aware loss functions are also promising directions to enhance robustness and clinical utility. Ultimately, this work provides a principled framework for adaptive multi-scale feature learning, with potential applicability to other medical image segmentation tasks characterized by high anatomical or pathological size variance.

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